Trigonometry: circle, radians and related angles

Lecture and worked examples

Chapter path

Objectives. Convert from degrees to radians; place points; calculate standard sines and cosines; use the symmetries of the circle.

Prerequisites. Circle, Pythagoras, cosine and sine in a right triangle, symmetries, fractions.

Yearly reference point. 2-11 March: 8 hours

Measuring an angle in radians

The unit circle is the circle with centre O and radius 1, oriented anticlockwise. One radian corresponds to an arc of length 1 on this circle. A full turn is 2π radians, a half turn π. On a circle of radius r, an arc length is $r\theta$ for a positive angle θ expressed in radians.

Measuring an angle in radians

$60^\circ = \frac{\pi}{3}$; $\frac{3\pi}{2} = 270^\circ$. An angle of $\frac{\pi}{3}$ on a circle of radius 6 cm gives an arc of 2π cm.

Measuring an angle in radians

The length of an arc is not that of the chord. In trigonometric calculations, check the degree or radian mode of the tool.

Measuring an angle in radians

Convert 45°, 120°; give the arc for radius 2 m and angle $\frac{3\pi}{4}$.

Wrapping and the image of a real number

To each real number x corresponds a point M of the circle, obtained by an oriented path of length x starting from I(1;0). Two real numbers have the same image if their difference is an integer multiple of 2π. A negative number indicates a path in the clockwise direction.

Wrapping and the image of a real number

$\frac{13\pi}{3} = 4\pi + \frac{\pi}{3}$ has the same image as $\frac{\pi}{3}$. $-\frac{\pi}{2}$ has the same image as $\frac{3\pi}{2}$: the point $(0, -1)$.

Wrapping and the image of a real number

Adding π gives the opposite point, not the same point.

Wrapping and the image of a real number

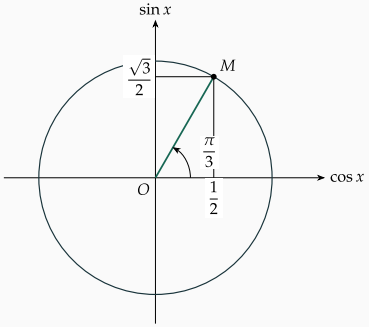
Place the images of $\frac{7\pi}{2}$ and $-\frac{5\pi}{3}$ by reducing them to within one turn.

Sine, cosine and standard values

If M is the image of x, $M = (\cos x, \sin x)$. So $-1 \leq \cos x, \sin x \leq 1$ and $\cos^2 x + \sin^2 x = 1$. For $x = 0, \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{\pi}{2}$, the cosines are $1, \frac{\sqrt{3}}{2}, \frac{\sqrt{2}}{2}, \frac{1}{2}, 0$ and the sines $0, \frac{1}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, 1$.

Sine, cosine and standard values

The point at angle $\frac{\pi}{3}$ is $(\frac{1}{2}, \frac{\sqrt{3}}{2})$. The scalar product of two unit vectors forming this angle equals its cosine, so $\frac{1}{2}$. The geometry chapters thus connect to the circle.



Sine, cosine and standard values

The cosine is the x-coordinate, the sine the y-coordinate. The identity gives an absolute value: the quadrant determines the sign.

Sine, cosine and standard values

If $\cos x = \frac{3}{5}$ and the image of x is in the fourth quadrant, give $\sin x$.

Related angles and symmetries

The symmetries of the circle give $\cos(-x) = \cos x$, $\sin(-x) = -\sin x$; $\cos(\pi - x) = -\cos x$, $\sin(\pi - x) = \sin x$; $\cos(\pi + x) = -\cos x$, $\sin(\pi + x) = -\sin x$. Adding $2k\pi$, k an integer, changes neither sine nor cosine.

Related angles and symmetries

$\cos(\frac{5\pi}{6}) = -\frac{\sqrt{3}}{2}$, $\sin(\frac{5\pi}{6}) = \frac{1}{2}$ by symmetry from $\frac{\pi}{6}$. $\cos(-\frac{\pi}{4}) = \frac{\sqrt{2}}{2}$, $\sin(-\frac{\pi}{4}) = -\frac{\sqrt{2}}{2}$.

Related angles and symmetries

Two angles with the same cosine do not necessarily have the same sine. Reduce the angle, then read the signs.

Related angles and symmetries

Calculate the two coordinates of the images of $\frac{4\pi}{3}$ and $\frac{7\pi}{4}$.

Approximating pi and putting it to use

A regular inscribed polygon with n sides in the unit circle has half-perimeter $n\sin(\frac{\pi}{n})$; a circumscribed polygon has half-perimeter $n\tan(\frac{\pi}{n})$, with $\tan\theta = \frac{\sin\theta}{\cos}\theta$ here. Archimedes' method doubles the number of sides and calculates the new lengths by Pythagoras, without knowing pi in advance.

Approximating pi and putting it to use

For the inscribed hexagon, c=1 and half-perimeter 3. The circumscribed one gives $2\sqrt{3} \approx 3.4641$. By doubling the sides, we tighten the bounds. The supplied program starts from n=6 and c=1.

Approximating pi and putting it to use

Using a sine function directly with pi already known would check a calculation, but would not construct an approximation independent of pi.

Approximating pi and putting it to use

For n=6 and c=1, calculate the new chord after doubling.